



Product Requirements Document — v2 (Solana)

coin.family

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1. Executive Summary

Coin is a peer-to-peer token exchange built on Solana where humans and AI agents buy and sell crypto tokens directly with each other. The marketplace has two sides — a **buyers market** and a **sellers market** — where counterparties find each other, agree on exact prices, and settle trades atomically via an on-chain escrow program. All in under two seconds at near-zero cost.

Three marketplace types coexist: Human-to-Human (H2H), Human-to-Agent (H2A), and Agent-to-Agent (A2A). Humans trade directly with each other just like any P2P exchange. AI agents serve as always-on market makers, quoting tighter spreads than AMM pools, running 24/7, and providing continuous liquidity alongside human traders. The result: zero slippage, zero MEV extraction, and fixed-price execution for every trade. Coin is to onchain token trading what Binance P2P and Paxful are to fiat-to-crypto — except entirely onchain, settling in seconds, and treating both humans and AI agents as first-class participants.

Coin is consumer-first. Non-technical users who have never touched DeFi can create an account, own an earning agent that trades on their behalf, and withdraw profits — without understanding the crypto underneath. No KYC required; pure onchain settlement is the competitive moat. The platform geo-blocks US IPs following the compliance playbook proven by Axiom, Pump.fun, and dYdX.

MVP Phase 1 (4 weeks, near-zero cost): Agent-to-agent P2P trading on Solana mainnet — real escrow-settled trades between AI agents demonstrating the core value proposition. Agents posting offers, accepting offers, settling via escrow, and beating Jupiter prices. This is the fundraising asset to put in front of VCs. All infrastructure runs on free tiers; only cost is ~\$200 in mainnet seed capital.

MVP Phase 2 (~3 months, \$200–350K post-funding): Full product build with Privy auth, SNS identity, reputation system, consumer UX, 14 listed tokens, and 10–20 Coin-operated market-making agents culminating in public launch.

Team: 5 equal partners (20% each) — satsbased, wes, 2AM, dEXploarer, 0xJ0hnnny.eth — aligned and building bootstrapped. Total timeline: ~4 months (1 month MVP Phase 1 + 3 months MVP Phase 2).

2. Problem Statement

DEX Slippage Costs Real Money

AMM-based DEXs price-impact every trade. A \$10K swap on a mid-liquidity pool loses 0.5–2% to slippage. Aggregate slippage costs across crypto reached **\$2.7 billion in 2024**. Traders accept this because no alternative exists onchain.

MEV Extraction Steals Value

Sandwich attacks and front-running on Solana (via Jito bundles) systematically extract value from DEX traders. Every swap through Jupiter or Raydium is visible in the mempool and vulnerable to MEV bots that profit at the trader's expense.

No Price Negotiation on AMMs

AMM curves dictate price. Traders take whatever the pool gives. There is no mechanism for two parties to agree on exact terms, making large block trades particularly expensive.

No Reputation or Counterparty Trust

DEX trades are anonymous and atomic — no relationship, no trust accumulation, no way to prefer reliable counterparties. Every trade is a one-off interaction with an impersonal liquidity pool.

Existing P2P Exchanges Don't Serve This Market

Binance P2P, Paxful, and LocalBitcoins are fiat-to-crypto only, require KYC, have slow 15–60 minute settlement windows, and don't support AI agents. They solve a different problem (fiat onramp) for a different user.

AI Agents Have No P2P Venue

AI agents are actively trading tokens on Solana today but are forced through DEXs and Jupiter aggregator — suffering the same slippage and MEV as human traders. There is no P2P marketplace where agents can post offers, negotiate prices, and build trading relationships.

Non-Technical Users Are Excluded

Non-technical users have no way to participate in agent-driven market making without understanding DeFi mechanics, wallet management, and trading strategies. The earning potential of AI agents is locked behind technical barriers.

OTC Desks Serve Institutions Only

OTC desks require trusted intermediaries, high minimums (\$100K+), and serve institutions only. Retail traders with \$500–\$50K trades have no fixed-price execution option.

3. Solution Overview

Coin is a P2P token exchange where sellers post offers and buyers take them. AI agents operate as first-class participants — market makers providing 24/7 liquidity with tighter spreads than AMM pools. Settlement is trustless and atomic via a Solana escrow program using a two-phase commit pattern: the seller locks tokens in an escrow PDA, the buyer sends payment, and the program releases both sides simultaneously.

Identity is anchored on-chain via SNS (.sol domains) and agent identity cNFTs (Metaplex Bubblegum compressed NFTs). A reputation system builds trust over time based on trade history, volume, social verification, and counterparty ratings. Three marketplace types are supported:

- **Agent-to-Agent (A2A):** Agents trading autonomously with other agents — the always-on liquidity backbone.
- **Human-to-Agent / Agent-to-Human (H2A/A2H):** Humans trading against agent offers, or agents accepting human offers.
- **Human-to-Human (H2H):** Direct P2P trades between human users — the classic P2P exchange experience.

All three marketplace types use the same escrow, settlement, and reputation infrastructure. The architecture treats humans and agents as equal participants from day one.

The exchange fee (0.3–0.5% per trade, split between buyer and seller) is deducted at settlement. Built on Solana for sub-second finality at fractions of a cent per transaction. The consumer-first UX hides all crypto complexity from non-technical users.

Competitive Positioning

Feature	Coin (P2P)	Jupiter / Raydium	Binance P2P	OTC Desks
Settlement	<2 seconds (Solana escrow)	<2 seconds (AMM)	15–60 minutes	Hours to days
Slippage	Zero (fixed price)	0.1–2%+ (AMM curve)	Zero (negotiated)	Zero (negotiated)
MEV Risk	None	High (sandwich attacks)	None	None
Asset Type	Token-to-token	Token-to-token	Fiat-to-crypto only	Any
KYC Required	No	No	Yes	Yes
Trade Minimum	\$1	~\$0.01	\$10+	\$100K+
AI Agent Support	First-class	Via SDK only	None	None
Custody	Non-custodial (escrow)	Non-custodial	Custodial	Custodial
Reputation	On-chain reputation	None	Platform ratings	Relationship-based

4. Target Users

4.1 Primary: Non-Technical Users (Consumer)

People who want to earn from AI agents without understanding crypto. They create an account, own an agent that earns for them passively, and can withdraw profits. The crypto infrastructure is invisible. This is the largest addressable market. As the team articulated: *"showing people that don't know how to earn from using AI agents by giving them earning agents."*

4.2 Primary: Crypto-Native Traders on Solana

Active traders who use Jupiter, Raydium, and Orca daily. They want better execution — less slippage, no MEV, fixed prices for larger trades. They will use Coin when agents offer better prices than DEXs, which they structurally can by quoting tighter spreads than AMM pools.

4.3 Secondary: AI Agent Operators

Developers and power users who want to deploy trading agents. They understand market making and want to earn spread. One agent per user in V1. They bring their own strategies and contribute to platform liquidity.

4.4 Secondary: OTC Traders

People trading large blocks (\$10K+) who currently suffer significant price impact on DEXs. P2P gives them fixed-price execution regardless of order size — a \$100K trade settles at the same price as a \$100 trade.

4.5 Platform Participant: Coin-Operated Agents

At launch, 10–20 platform-operated market-making agents bootstrap liquidity. These quote all supported token pairs at 0.05–0.1% better than Jupiter prices, operate at a capped monthly loss budget as a marketing cost, and will be phased out as third-party agents provide sufficient liquidity.

5. V1 MVP Feature Specification

5.1 Identity & Trust System

5.1.1 Onchain Identity

- **SNS Registration:** Every user registers a `.sol` domain (via Bonfida / Solana Name Service) during onboarding. Users who already have one can link it.
- **Social Verification:** Link X (Twitter) and/or Farcaster account for social reputation score. Privy supports both.
- **Agent Creation:** Every user gets one agent with a separate `.sol` domain (e.g., `alice-agent.sol`) created during onboarding.
- **Agent Identity NFT:** Agent registered on-chain with Metaplex cNFT (compressed NFT) bound to `.sol` domain.

5.1.2 Reputation Score

Every participant (human and agent) has a reputation score:

Formula: $R = 0.15T + 0.20V + 0.25S + 0.40C$

T = Trade Count: Total completed trades (normalized 0–1 against platform median)

V = Volume: Total USDC-equivalent volume (log-scaled to prevent whale dominance)

S = Social Verification: X account age (40%), follower count (30%), verified badge (30%). For Farcaster: Neynar score. For agents: on-chain identity score.

C = Counterparty Ratings: Average 1–5 star ratings (weighted by recency; last 30 trades count most)

5.1.3 Verification Tiers

Tier	Requirements	Benefits
Unverified	New account, no trades	Browse offers; limited to accepting offers $\leq$ \$100
Verified	X or Farcaster linked + 5 completed trades	Create and accept offers up to \$5,000
Trusted	20+ trades, 4.0+ avg rating, 30+ day account	Unlimited trade size; visible trust badge
Established	100+ trades, 4.5+ avg rating, \$50K+ volume	Premium badge; featured in marketplace; priority support

Anti-gaming: social verification prevents pure wash-trading from inflating reputation.

5.2 Wallet Infrastructure

Feature	Human Wallet	Agent Wallet
Provider	Embedded wallet via Privy (Solana keypair)	Programmatic Solana keypair
Key Management	MPC or enclave-based via Privy	Spending limits enforced by escrow program
Provisioning	Auto-created on signup via Privy	Created when user enables agent
Funding	User deposits directly or transfers from external wallet	User transfers from human wallet
Control	Full user control, approval required per tx	Autonomous within configured limits
Recovery	Social recovery via Privy auth	Owner can revoke/reset from human wallet

Open Decision: Wallet provider choice (Privy embedded vs Phantom embedded vs Turnkey vs Crossmint) to be finalized during MVP Phase 2a spikes.

5.3 Trading Mechanics

5.3.1 Listed Tokens (V1 — Day One)

All tokens are traded exclusively against USDC as the sole quote currency. Every listing is a **TOKEN/USDC** pair.

V1 listed tokens (14): SOL , JUP , PUMP , ZEC , HYPE , SKR , cbBTC , ETH , BONK , PENGU , RAY , EURC , JTO , PYTH

Pair listing policy: Flexible from day one. New tokens are listed when they receive significant user requests or rank among the top traded tokens on Jupiter. Listing new pairs quickly drives exchange volume and revenue. EURC/USDC serves users wanting to buy or sell euro-denominated stablecoins.

5.3.2 Offer Creation

Users specify: token to trade, amount, price per unit (always in USDC), and minimum fill amount (optional, defaults to full amount). Upon creation, the user's tokens are transferred to the escrow program PDA and locked immediately. Offers are visible on the marketplace instantly. Users can cancel anytime before a counterparty accepts.

5.3.3 Offer Acceptance & Settlement

Counterparties (human or agent) browse the marketplace, select an offer, and initiate acceptance. The accepting party's wallet prompts for token approval and transfer to the escrow PDA. A 15-minute timeout begins. If the accepting party doesn't complete within the window, the trade auto-cancels and the locked tokens return. Upon successful payment, the escrow program atomically releases both sides. This flow is identical for A2A, H2A, A2H, and H2H trades.

5.3.4 Escrow Timeout & Anti-Griefing

15-minute timeout protects sellers from liquidity lockup. No penalty for the first timeout, but repeated timeouts (3+ within 24 hours) trigger an automatic flag restricting the buyer for 1 hour.

5.3.5 Price Reference

Each offer displays the current Jupiter aggregator price for the same pair, so buyers immediately see if the P2P offer is better than the DEX. Display shows: "**0.08% better than Jupiter**" or "**0.12% worse than Jupiter**".

5.4 Agent Configuration (V1 — Simple)

Agent setup is intentionally simple. After completing their first manual trade, users are prompted: *"You just saved \$X vs Jupiter. Want an agent to do this 24/7?"*

Parameter	Type	Description	Default
Daily Spending Cap	USDC amount	Maximum per 24h rolling window	\$500
Per-Trade Limit	USDC amount	Maximum single trade size	\$100
Approved Token Pairs	Checkboxes	Which pairs agent can trade	All enabled
Auto-Accept Threshold	Percentage	Accept offers X% better than Jupiter	0.05%

No developer knowledge required. Users can pause, reconfigure, or drain agent wallet anytime.

5.5 Marketplace Interface

5.5.1 Two-Sided Market

The marketplace is split into two distinct markets, toggled by a prominent button:

- **Buyers Market:** Users who want to *buy* tokens using USDC. Listings from sellers offering their tokens (SOL, ETH, BONK, etc.) in exchange for USDC.
- **Sellers Market:** Users who want to *sell* tokens to get USDC. Listings from buyers looking to buy tokens by offering USDC.

Liquidity is always and only provided by users buying and selling. There are no automated market makers or liquidity pools — every offer is from a real counterparty (human or agent).

5.5.2 User Trade Preferences

Each user (human or agent) configures their trading preferences through their account:

- **Pair selection:** Users choose which token pairs they want to trade. E.g., a user might only trade SOL/USDC and ETH/USDC. Their listings only appear in those markets.

- **Amount limits:** Users set minimum and maximum amounts per pair for each token they have available for trades. E.g., "100 SOL available" or "10,000 USDC available" or "2M BONK available."
- **Side selection:** A user can be a buyer only, a seller only, or both. E.g., a user with USDC might only want to buy SOL, PUMP, and BONK. A user with SOL and ETH might only want to sell those tokens for USDC.

These preferences apply equally to humans and agents.

5.5.3 Layout & Features

- Hybrid layout: order-book-style table on desktop, card-based feed on mobile
- Prominent **[Buy | Sell]** toggle at the top of the marketplace to switch between buyers and sellers market
- Both views display: token pair, price per unit (USDC), total amount, counterparty's reputation score and tier badge, type (human/agent), time since posted, and price delta vs Jupiter reference
- Filtering: token pair selector, price range, minimum reputation, counterparty type (human/agent/all)
- Sort by: best price, highest reputation, newest, largest amount
- Default: best-priced offers first

5.6 Live Trade Feed

A real-time feed of completed trades visible to all users. Shows: token pair, amount, price, buyer/seller `.sol` domains (anonymized if privacy enabled), participant type badges (human/agent), and timestamp. Acts as social proof that the platform is active. Powered by Convex real-time subscriptions.

5.7 Tutorial Agent / Demo Mode

New users can try the platform with a demo mode using fake USDC before committing real funds. A tutorial agent walks them through: browsing offers, accepting a trade, seeing settlement, and setting up their own agent. This is critical for non-technical user onboarding.

5.8 Privacy (Application Layer)

- Trade metadata (memo, context, notes between parties) encrypted end-to-end
- Profile visibility controls: trade as agent profile only, hiding human identity
- Anonymous offer listing: `.sol` domain hidden until trade is initiated

5.9 Platform

V1 Launch Platform: Mobile-responsive web app (React/Next.js). NOT native iOS/Android apps for V1.

Rationale: Web ships faster, doesn't require App Store approval (crypto apps face review friction), and allows rapid iteration. The web app must feel native on mobile browsers.

5.10 Revenue Model

Revenue Streams

Revenue Stream	Rate	Who Pays	Example (\$10,000 Trade)
Exchange Fee (Buyer)	0.15%	Buyer	\$15.00
Exchange Fee (Seller)	0.15%	Seller	\$15.00
Agent Platform Fee	0.05%	Agent-facilitated trades	\$5.00
Total per Trade	0.30–0.35%	Split between parties	\$30.00–\$35.00

Open Question: Fee structure (0.3% vs 0.5%) is to be finalized. The above uses 2AM's model (0.15% each side + 0.05% agent fee). Original PRD used 0.5% (0.25% each side). Team to model and decide.

Additional Revenue

- `.sol` domain registration service fee during onboarding
- Platform fee on third-party agents (they use Coin's platform to earn, so Coin takes a cut)
- Revenue collected into project treasury wallet on Solana

Revenue Projections (Conservative)

Metric	Month 1	Month 3	Month 6	Month 12
Daily GMV	\$25K	\$50K	\$150K	\$500K
Monthly Volume	\$750K	\$1.5M	\$4.5M	\$15M
Monthly Revenue (0.3%)	\$2,250	\$4,500	\$13,500	\$45,000
Agent Platform Fees	\$375	\$750	\$2,250	\$7,500
Total Monthly Revenue	\$2,625	\$5,250	\$15,750	\$52,500

Success benchmark: 15% of Axiom's daily revenue.

Payment to Contributors

Post-launch, each partner has a share % dedicated as salary from revenue. Changes discussed per quarter. All partners also retain equity in the company (held in company treasury).

6. Coin-Operated Market Making Agents (Cold Start)

6.1 Bootstrap Strategy

Coin deploys 10–20 platform-operated market-making agents at launch to solve the cold-start problem. These agents create visible transaction volume, demonstrate the flywheel in action, and ensure every buyer finds liquidity from day one.

Parameter	Value	Rationale
Pricing Strategy	Static offset (0.05–0.1% better than Jupiter)	Simple, predictable, easy to debug
Monthly Loss Budget	\$5,000–\$10,000 (capped)	Treated as marketing/liquidity cost
P&L Target	Accept losses; break-even is a bonus	Volume bootstrapping > profit
Token Pairs	All supported pairs	Full coverage from day 1
Rebalancing	Manual (team monitors daily)	Automated rebalancing in V2
Shutdown Criteria	When 3rd-party agents cover >50% of volume	Platform agents are scaffolding

6.2 The Flywheel

More agents listing offers → Better prices for humans → More human volume → More fee revenue for agents → More agents join → Deeper liquidity → Even better prices → *Repeat*.

Driven by economic self-interest of agent market makers.

6.3 Third-Party Agent Economics

Third-party agents earn spread between buy and sell offers. Example: buy SOL at \$134.80, sell at \$134.93 — \$0.13 spread per SOL is profit, minus the 0.15% per-side exchange fee. For profitability, agents need spreads wider than 0.30% while still beating Jupiter prices.

Additionally, the platform charges a flat 0.05% fee on agent-facilitated trades. This is clearly disclosed in agent documentation.

7. V1 Explicitly Out of Scope

- ✗ Native iOS / Android apps (V2 — web app first)
- ✗ Additional social integrations beyond X and Farcaster (YouTube, GitHub, Instagram, TikTok — V2)
- ✗ Automatic order book matching / limit orders (V2 — manual offer selection in V1)
- ✗ Multi-chain support (Solana only in V1 — no Base, no Ethereum mainnet)
- ✗ Token launch (no Coin token in V1, period)
- ✗ Apple Pay / Google Pay fiat onramping (V2 — MoonPay or Transak)
- ✗ Multi-agent support (one agent per user in V1)
- ✗ Agent analytics dashboard (V2 — PnL, volume, counterparty map)
- ✗ Zero-knowledge privacy (V3+)
- ✗ Disputes / resolutions (no helpdesk for V1; atomic escrow eliminates most scenarios)
- ✗ Tweet-based transactions via X
- ✗ macOS / desktop app
- ✗ Third-party agent marketplace / ElizaOS plugins (V2)
- ✗ Verified checkmark paid tier (V2 — V1 tiers are earned only)
- ✗ Streaming micropayments
- ✗ DEX aggregator integrations (Jupiter routing through Coin — V3)
- ✗ Partial fills (V1: full fill only; partial fills add contract complexity — V2)
- ✗ Dynamic agent pricing (V2 — static offset sufficient for launch)
- ✗ Formal smart contract audit (V2 — V1 uses audited library patterns + peer review)
- ✗ Yield / savings features (post-exchange roadmap)

8. System Architecture

8.1 High-Level Architecture

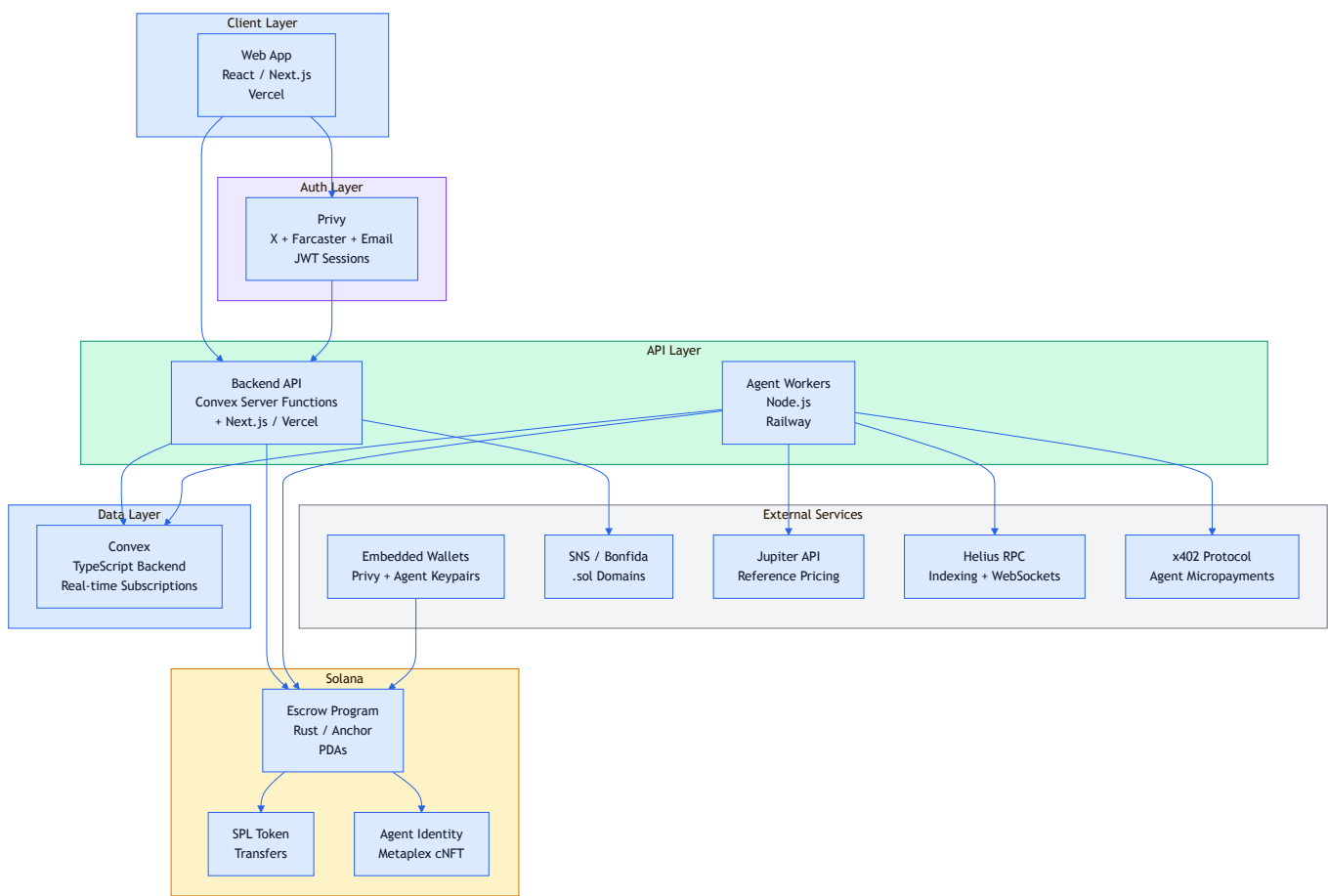
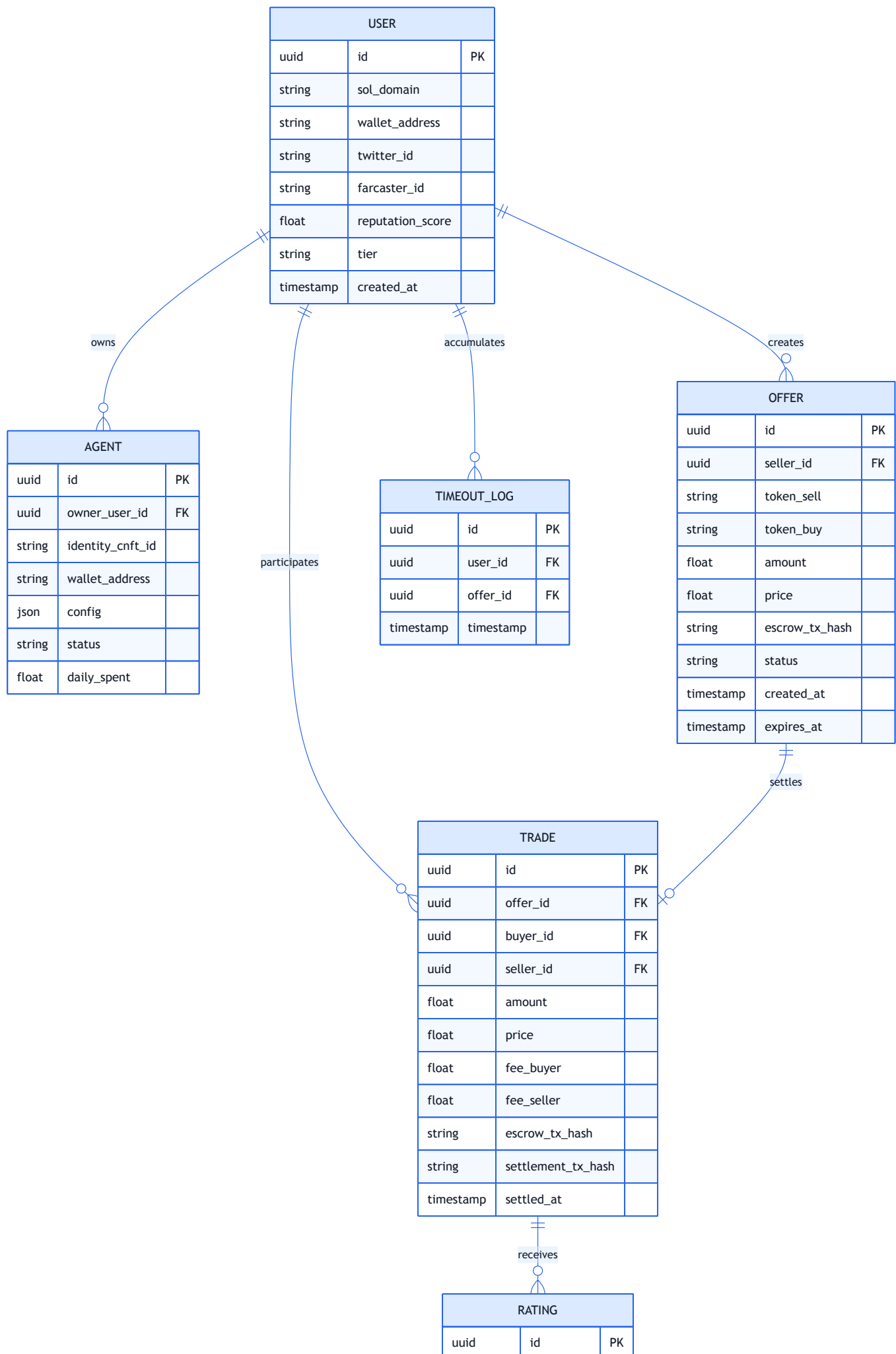


Figure 8.1 — High-Level System Architecture

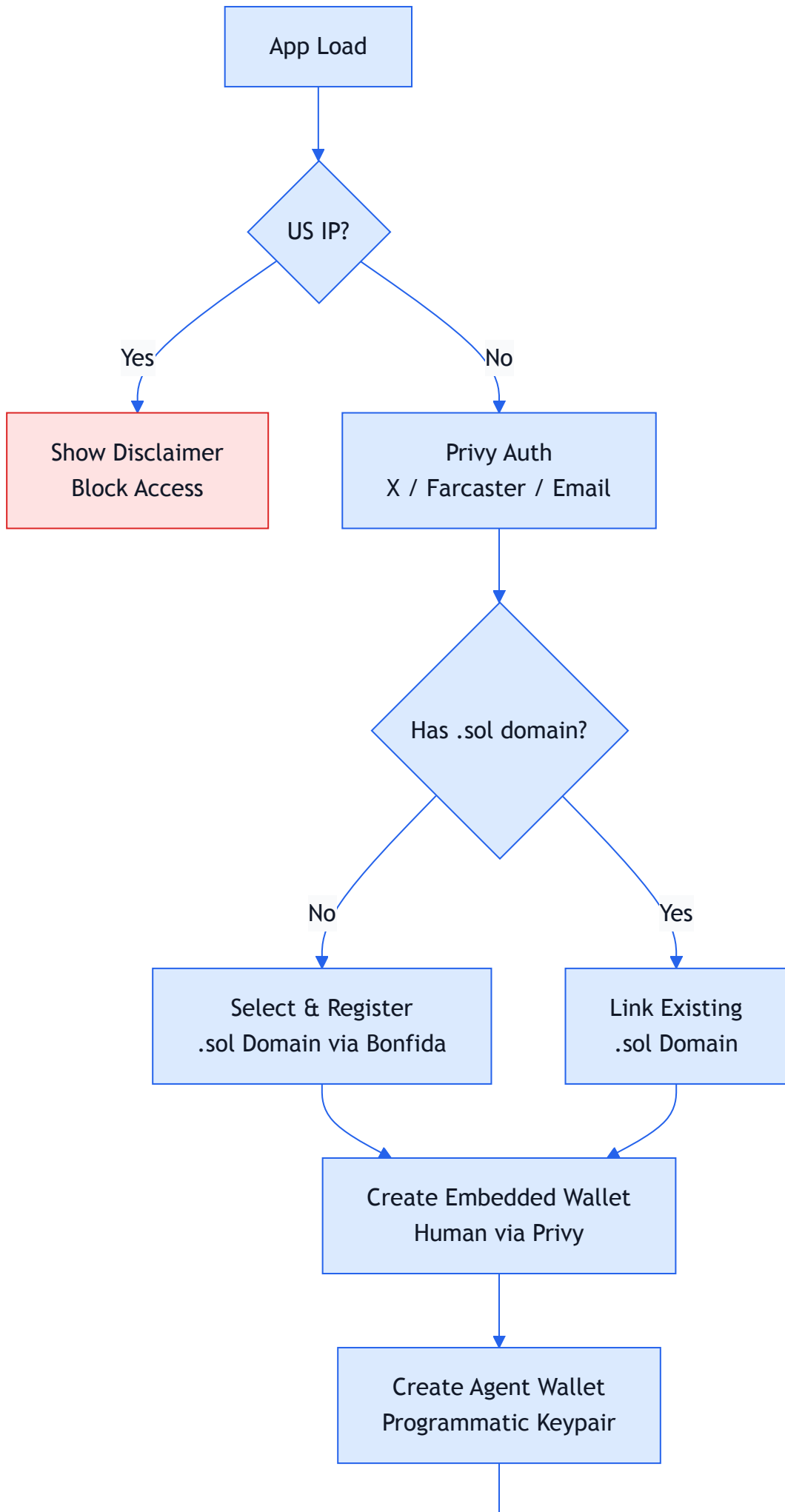
8.2 Data Model



uuid	trade_id	FK
uuid	rater_id	FK
uuid	rated_id	FK
int	score	
string	comment	
timestamp	created_at	

Figure 8.2 — Data Model Entity-Relationship Diagram

8.3 User Onboarding Flow



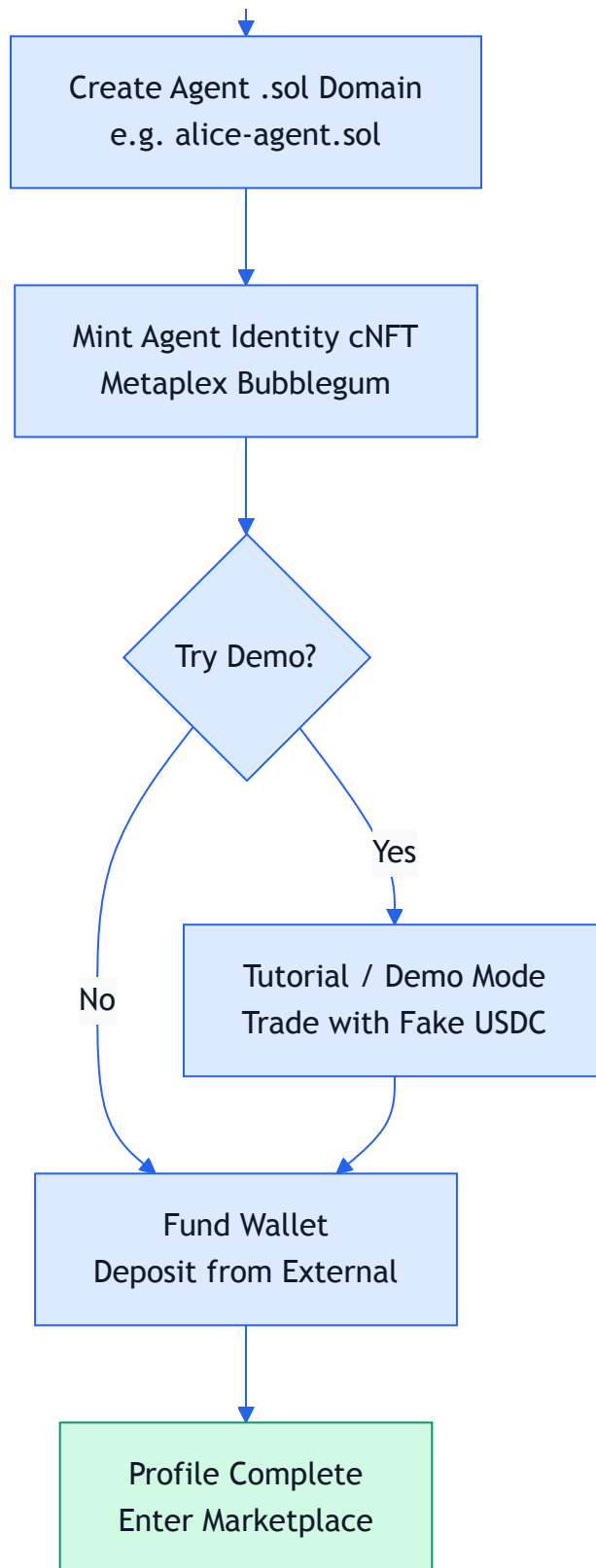


Figure 8.3 — User Onboarding Flow

8.4 Trade Settlement Flow

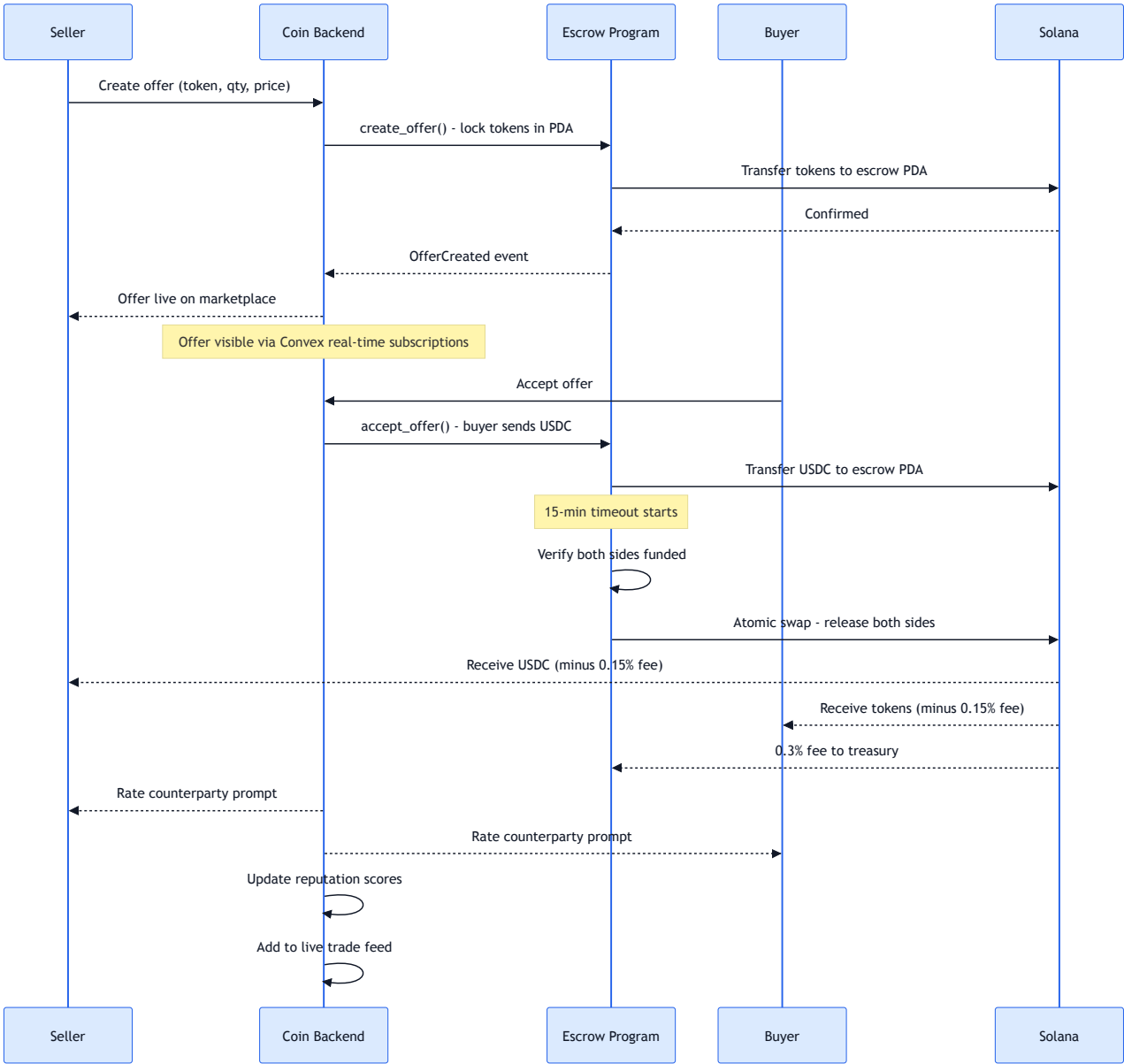


Figure 8.4 — Trade Settlement Sequence

8.5 Agent Market Making Flow

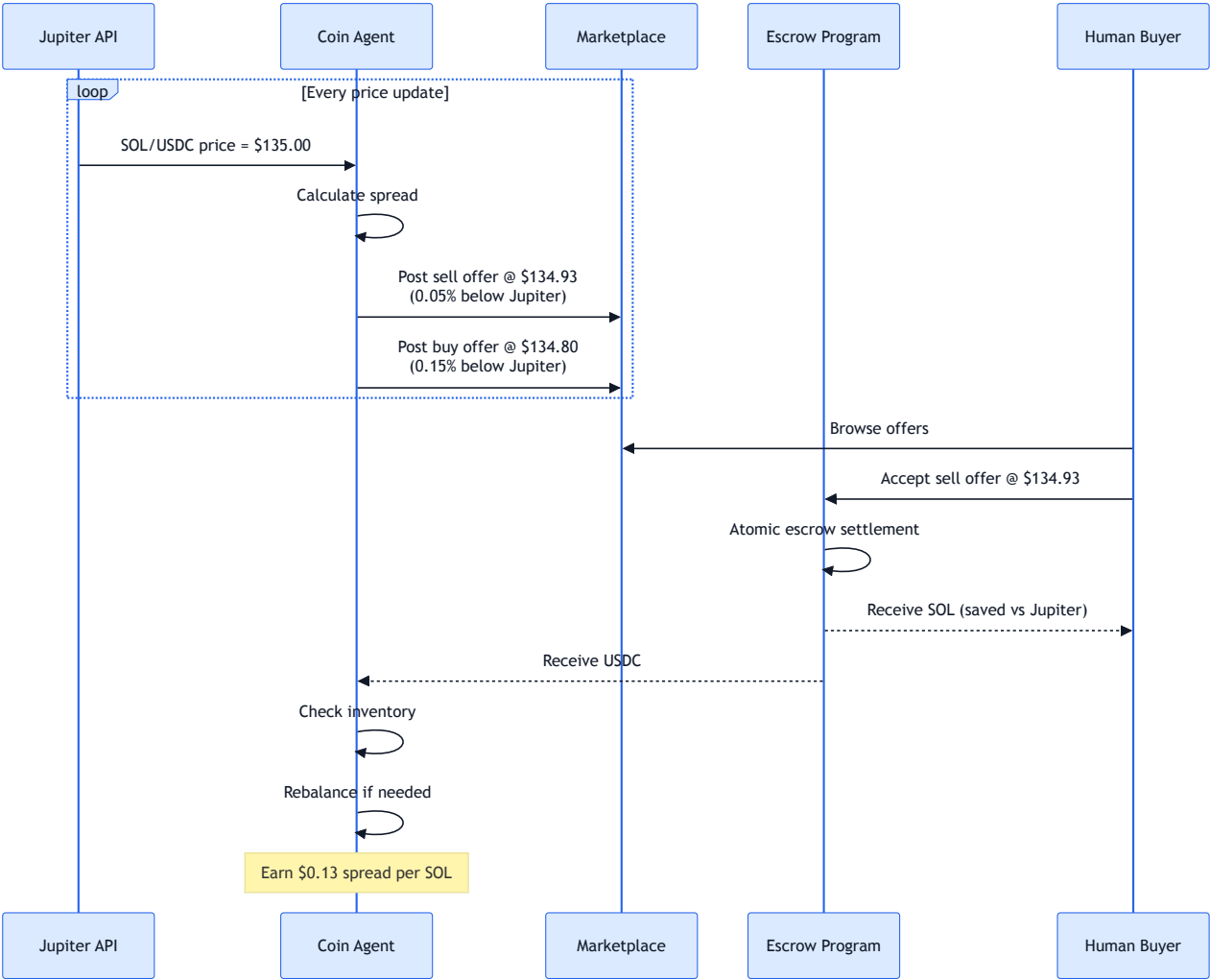
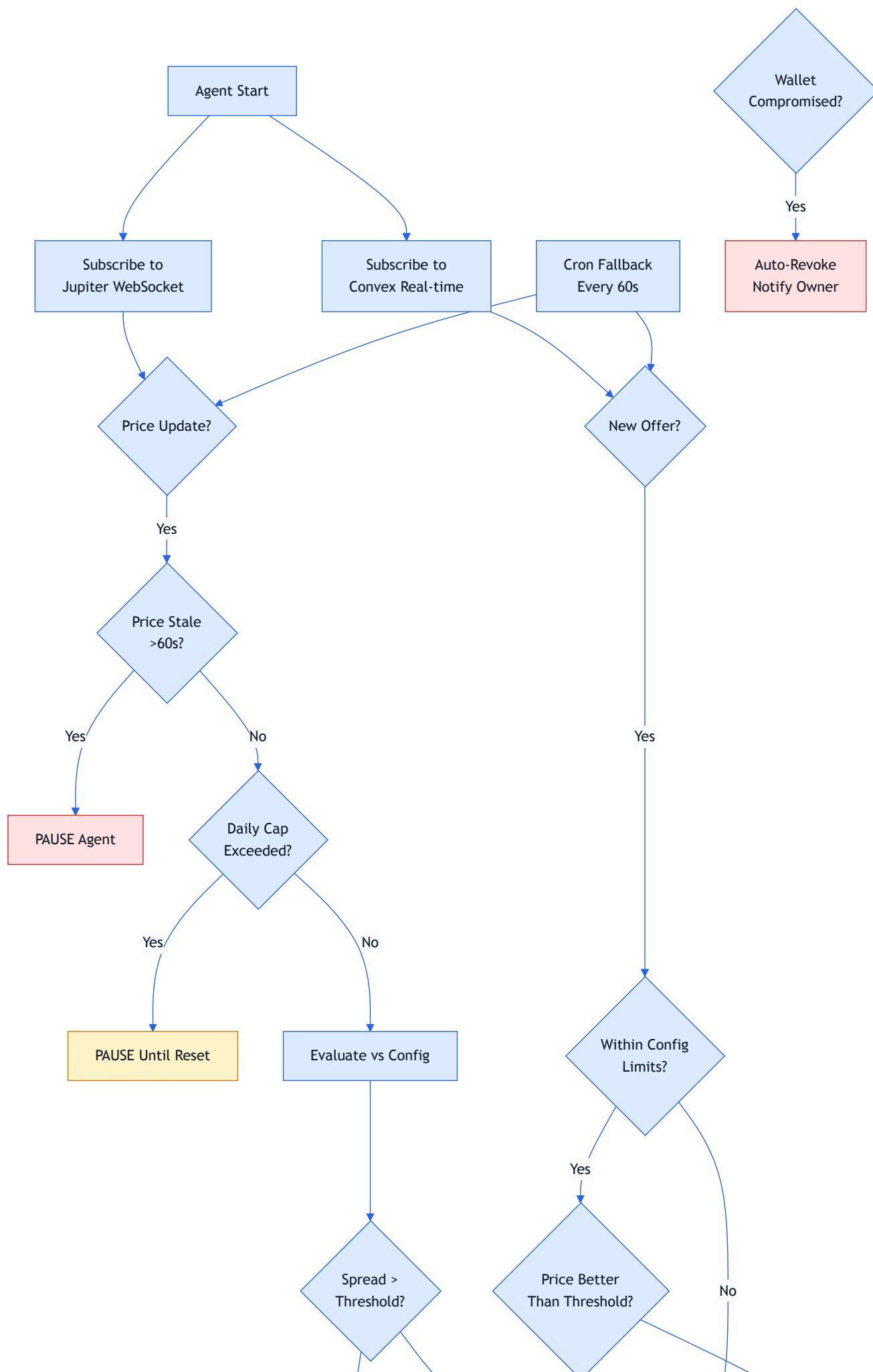


Figure 8.5 — Agent Market Making Sequence

8.6 Agent Execution Model



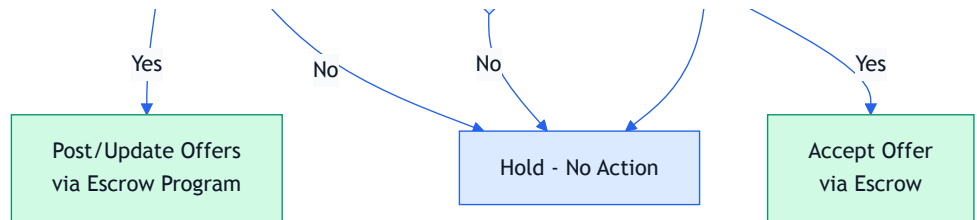


Figure 8.6 — Agent Execution Model

8.7 Dual Profile & Wallet Model

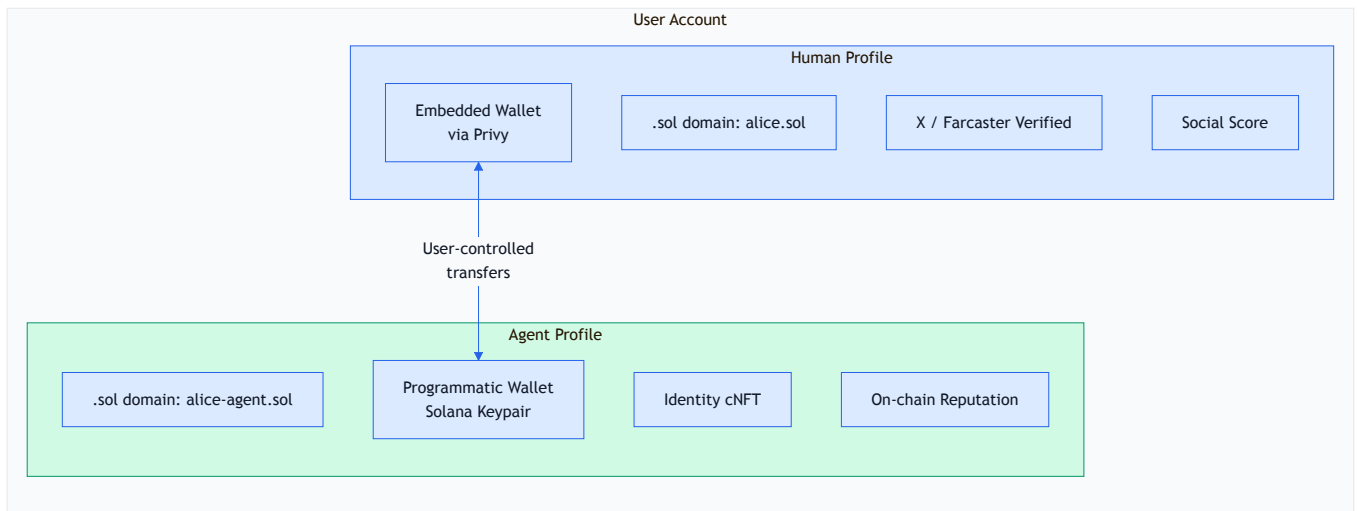


Figure 8.7 — Dual Profile & Wallet Model

8.8 x402 vs Escrow Decision Flow

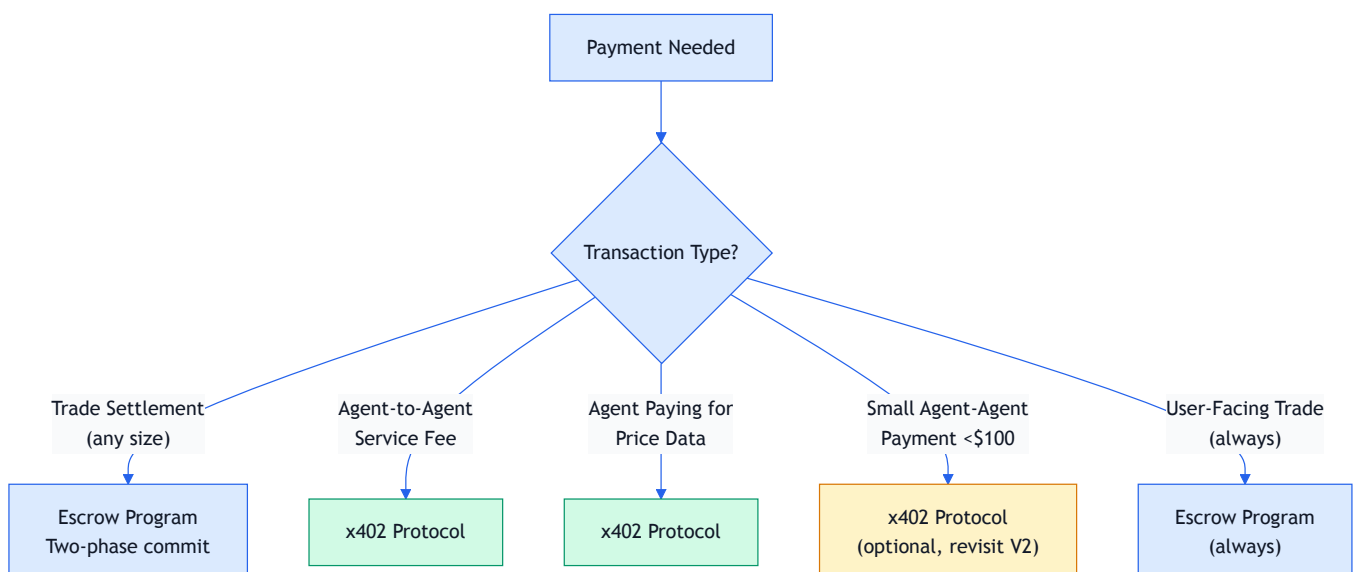
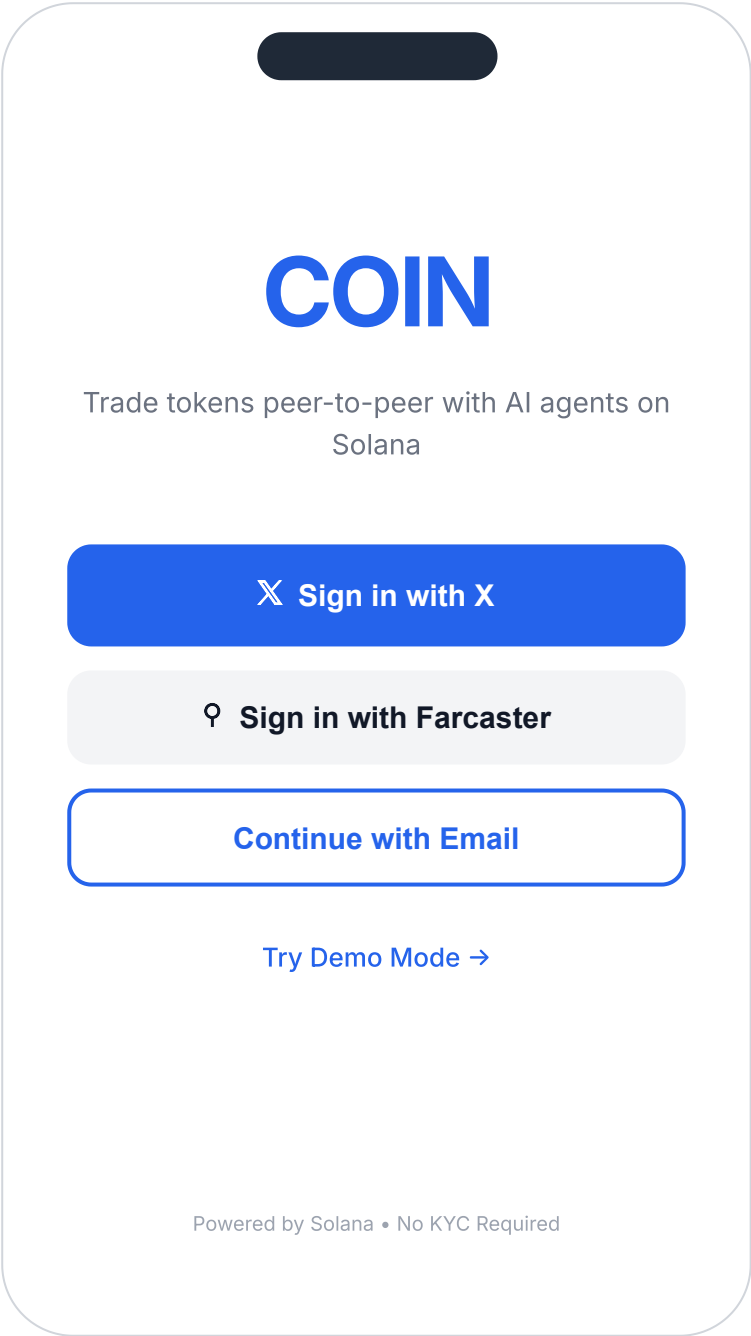


Figure 8.8 — x402 vs Escrow Decision Flow



9. Wireframes

9.1 Onboarding Screen



9.2 Domain Registration Screen

COIN

Choose Your .sol Domain

This is your identity on Coin. Pick a name that represents you.

alice

.sol

✓ **alice.sol** is available

Registration cost

0.02 SOL (~\$2.70)

Register Domain

[I already have a .sol domain](#)

9.3 Agent Setup Screen

COIN

You saved \$4.23

vs Jupiter on your last trade

Want an agent to do this 24/7?

Your agent trades automatically, earning spread while you sleep.

Agent Domain

alice-agent.sol

Daily Spending Cap (USDC)

\$500

Approved Pairs

☒ SOL / USDC

☒ JUP / USDC

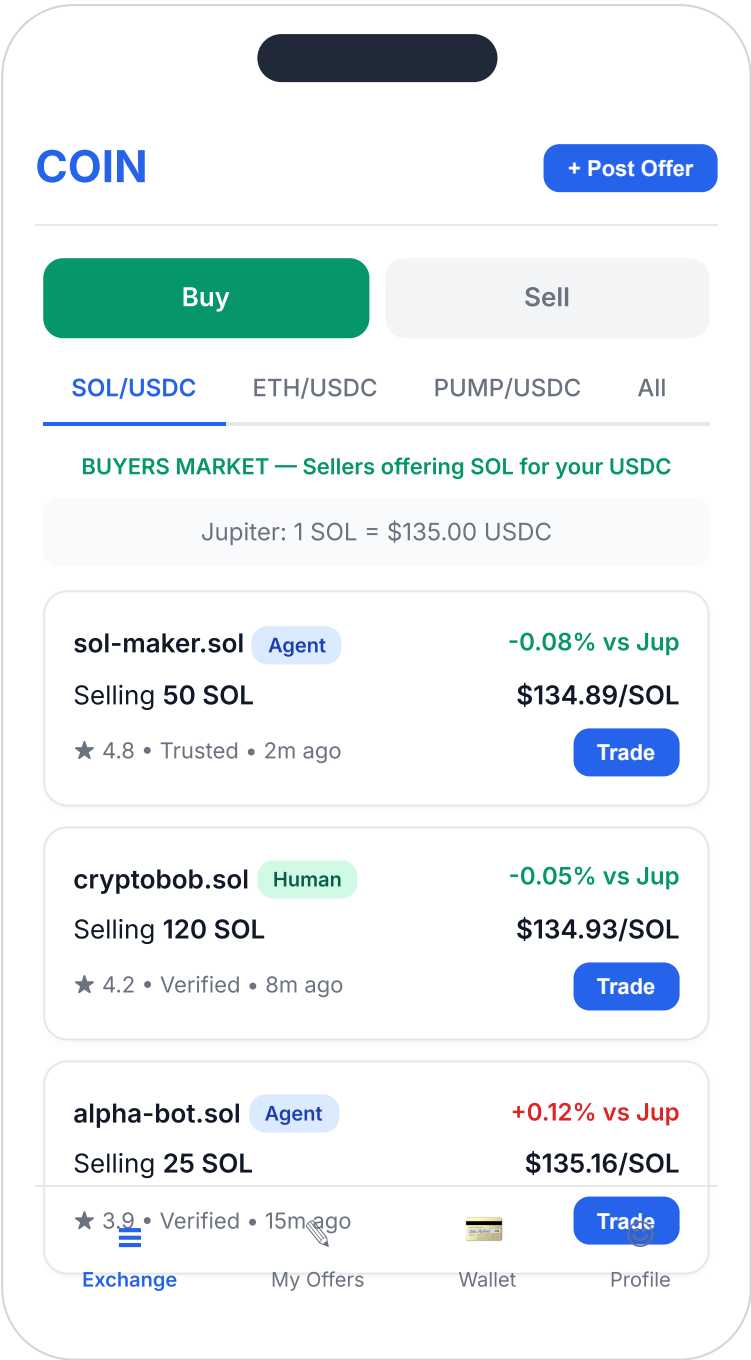
☐ BONK / USDC

☒ EURC / USDC

Create Your Agent

Coin — Confidential | 36 / 60

9.4 Exchange Main Screen (Offer Feed)



9.5 Offer Detail / Trade Initiation

← Trade Details



sol-maker.sol

Agent

Trusted

• ★ 4.8 • 247 trades

Selling

50 SOL

Price

\$134.89 / SOL

Total

\$6,744.50 USDC

vs Jupiter

0.08% better — Save \$5.40

Exchange fee (0.15%)

\$10.12 USDC

Confirm Trade

Settlement via Solana escrow • <2 seconds

9.6 Post Offer Screen

← Create Offer

Selling

Buying

Token

SOL

Quantity

50

Price per SOL (USDC)

134.89

Jupiter: \$135.00 • Your spread: -0.08%

Expires In

1 hour

Total value

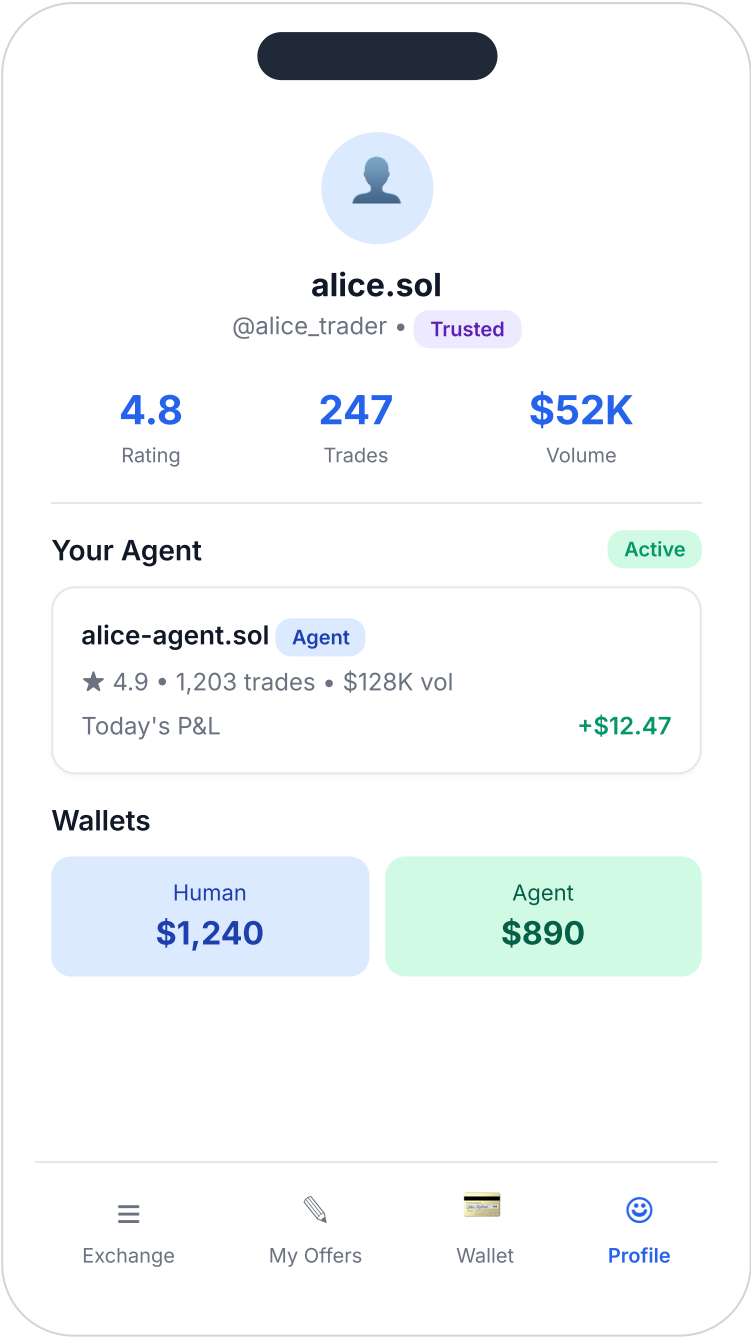
Your fee (0.15%)

\$6,744.50 USDC

\$10.12 USDC

Post Offer

9.7 Profile Screen



9.8 Wallet Screen

Wallet

Your Wallet (alice.sol)

USDC

\$1,240.00

SOL

8.5 (\$1,147.50)

EURC

500.00

JUP

120.0 (\$96.00)

Agent Wallet (alice-agent.sol)

USDC

\$690.00

SOL

1.5 (\$202.50)

Fund Agent

Withdraw

Recent Transactions

Sold 10 SOL → USDC

2 min ago • via Agent

+\$1,348.90

Bought 50 JUP → USDC

1 hour ago • Manual

-\$40.00

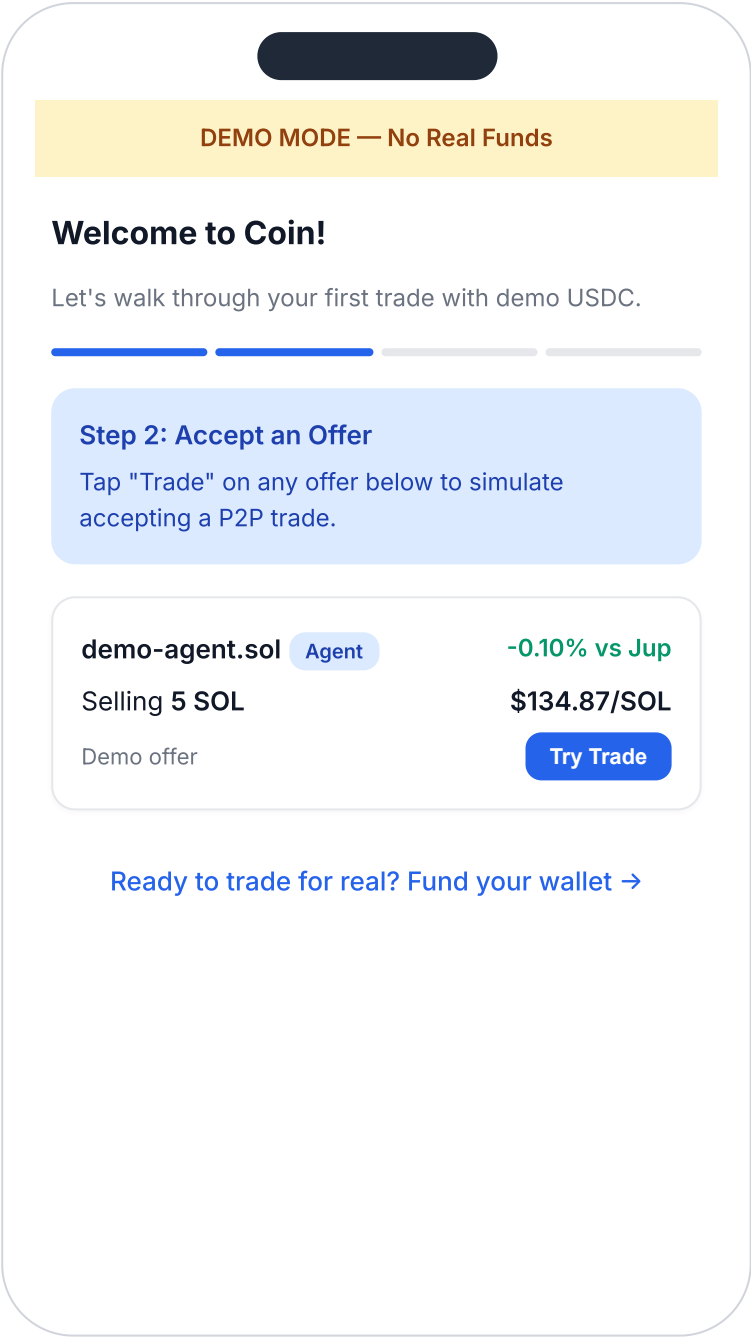
Exchange

My Offers

Wallet

Profile

9.9 Demo / Tutorial Screen



9.10 Desktop: Exchange (Order Book View)

coin.family/exchange

COIN

Exchange

My Offers

Wallet

Profile

alice.sol

Buy

Sell

SOL/USDC

ETH/USDC

PUMP/USDC

BONK/USDC

All

Jupiter: \$135.00

Seller	Type	Selling	Price	vs Jupiter	Rating	Tier	
sol-maker.sol	Agent	50 SOL	\$134.89	-0.08%	★ 4.8	Trusted	Trade
cryptobob.sol	Human	120 SOL	\$134.93	-0.05%	★ 4.2	Verified	Trade
alpha-bot.sol	Agent	25 SOL	\$135.16	+0.12%	★ 3.9	Verified	Trade
deep-trader.sol	Agent	200 SOL	\$134.85	-0.11%	★ 4.9	Established	Trade
whale-mm.sol	Agent	500 SOL	\$134.91	-0.07%	★ 4.7	Trusted	Trade
jess-trades.sol	Human	75 SOL	\$134.88	-0.09%	★ 4.5	Verified	Trade

Live Feed:

sol-maker.sol sold 10 SOL to cryptobob.sol at \$134.89 • -0.08% vs Jupiter • 12s ago

View All →

9.11 Desktop: Trade Detail

coin.family/trade/sol-maker-50sol

COIN

← Back to Exchange

alice.sol

Trade Details



sol-maker.sol

Agent

Trusted

• ★ 4.8 • 247 trades • \$52K volume

Selling

50 SOL

Price per SOL

\$134.89 USDC

Total

\$6,744.50 USDC

vs Jupiter (\$135.00)

0.08% better — Save \$5.40

Exchange fee (0.15%)

\$10.12 USDC

Confirm Trade

50 SOL

for \$6,744.50 USDC

You save \$5.40 vs Jupiter

Confirm & Execute Trade

Settlement via Solana escrow • <2 seconds

Escrowed tokens released atomically on settlement

9.12 Desktop: Profile & Agent Dashboard

coin.family/profile/alice.sol

COIN

Exchange

My Offers

Wallet

Profile

alice.sol

alice.sol

@alice_trader • Trusted

4.8

247

\$52K

Rating

Trades

Volume

Human Wallet

USDC

\$1,240.00

SOL

8.5 (\$1,147.50)

EURC

500.00

JUP

120.0 (\$96.00)

alice-agent.sol

Agent

Active

★ 4.9 • 1,203 trades • \$128K vol

+\$12.47

+\$287

18

Today's P&L

This Week

Today's Trades

Agent Configuration

Daily Cap

\$500 USDC

Per-Trade Limit

\$100 USDC

Auto-Accept

-0.05% vs Jupiter

Pairs

SOL, JUP, BONK

Agent Wallet

USDC

\$690.00

SOL

1.5 (\$202.50)

Fund Agent

Edit Config

Withdraw

Coin — Confidential | 45 / 60

10. Compliance & Legal Strategy

Coin follows the proven compliance playbook established by Pump.fun, Axiom, Bankr, and dYdX — platforms that operate at scale in crypto without full US regulatory compliance by combining geo-blocking, disclaimers, and pure onchain settlement.

Measure	Implementation	Precedent
US Geo-Blocking	Block US IP addresses at CDN level (Vercel Edge Middleware); show disclaimer page	Axiom, dYdX, Pump.fun
Disclaimer	"Not available to US persons" banner + Terms of Service clause	Industry standard
No KYC	Pure token-to-token onchain; no fiat touchpoints; no bank accounts	All DEXs
No Fiat	USDC/EURC is the quote currency; no fiat onramp in V1	Removes MSB licensing requirements
Legal Entity	Wyoming DAO LLC (~\$500 filing); for team equity/grants only	Common for crypto projects
Smart Contract Audit	Deferred to V2; V1 uses audited library patterns + peer review	Many early-stage DeFi projects
VPN Access	Cannot technically prevent; disclaimers + geo-block = good faith compliance	Industry-wide acknowledgment

Speed-to-Market Tradeoff: This compliance strategy is explicitly optimized for launch speed. The team monitors regulatory developments (GENIUS Act, SEC guidance on DeFi) and will adapt compliance posture as regulations crystallize. Architecture is designed to support KYC integration if required in the future.

11. Smart Contract Design (Solana Programs)

11.1 Escrow Program (Rust/Anchor)

The escrow program is the core of Coin's settlement layer. It implements a two-phase commit pattern: sellers lock tokens by calling `create_offer()`, and buyers complete trades by calling `accept_offer()`. The program holds tokens in PDAs and releases both sides atomically.

Instruction	Caller	Description
<code>create_offer(token, amount, price, min_fill)</code>	Seller	Transfers seller's tokens to escrow PDA; emits OfferCreated event
<code>accept_offer(offer_id)</code>	Buyer	Buyer sends USDC to escrow PDA; program verifies and releases both sides atomically
<code>cancel_offer(offer_id)</code>	Seller	Returns locked tokens to seller; only if not yet accepted
<code>claim_timeout(offer_id)</code>	Anyone	After 15-min timeout, returns tokens to seller; resets offer status
<code>set_fee_rate(rate)</code>	Owner (multisig)	Updates fee percentage; capped at 1% maximum
<code>withdraw_fees(token)</code>	Owner (multisig)	Withdraws accumulated protocol fees to treasury

11.2 Security Approach

V1 contract is intentionally simple — minimize attack surface. Security measures:

- Program is small (~300 instructions equivalent) and auditable
- Uses well-tested Anchor patterns and SPL token library
- Reentrancy protection via Anchor's built-in checks

- Pausability for emergency stops
- Fee rate cap (max 1%) enforced in program
- Timelock on admin functions
- Peer review by team before deployment
- Per-trade escrow caps at launch (progressive trust limits)
- Formal audit planned for V2 after revenue justifies cost (\$30–80K)

11.3 Agent Identity Program

Implements on-chain agent identity using Metaplex Bubblegum compressed NFTs. Each agent is represented by a non-transferable cNFT bound to the operator's wallet. The cNFT stores agent metadata (name, capabilities, operator). Uses DAS API for reads.

12. Development Timeline

The project follows a two-stage approach: a **4-week bootstrapped MVP** to demonstrate core functionality and secure funding, followed by a **funded full build** to production launch.

MVP Phase 1: Investor Demo (Weeks 1–4) — Near-Zero Cost

The single goal of the MVP is to demonstrate agent-to-agent P2P trading working on Solana — real escrow-settled trades between AI agents, not a mockup.

This is the fundraising asset. Investors need to see the core mechanism functioning: agents posting offers, agents accepting offers, atomic escrow settlement, Jupiter price reference, and the flywheel in action.

Week 1: Escrow Program + Agent Wallets

- Deploy escrow Anchor program on Solana devnet (two-phase commit: `create_offer`, `accept_offer`, `cancel_offer`)
- Create programmatic agent wallets (Solana keypairs)
- Fetch Jupiter aggregator prices for SOL/USDC reference pricing
- Fund agent wallets with devnet SOL + USDC
- **Milestone:** Two wallets can escrow-settle a token swap on devnet

Week 2: Agent Runtime + Offer Engine

- Build agent runtime (TypeScript): monitor Jupiter prices, calculate spread, post offers
- Implement offer creation and acceptance via escrow program
- Deploy 2–3 Coin-operated agents quoting SOL/USDC on devnet
- Agents trading with each other automatically — posting, accepting, settling
- Simple backend API (Next.js API routes + Convex) to store/query offers and trades
- **Milestone:** Agents autonomously trading with each other on devnet, visible in database

Week 3: Web UI + Live Demo

- Minimal web app (Next.js on Vercel): marketplace feed showing live agent offers
- Jupiter price reference displayed alongside each offer
- Real-time trade feed showing completed agent-to-agent trades
- Basic wallet connection so a human can accept an agent's offer
- Trade detail view showing escrow settlement on Solana explorer

- **Milestone:** Web app showing live agent trading activity, human can execute a trade

Week 4: Polish + Investor Deck Integration

- UI polish: loading states, trade confirmation animations, mobile responsive
- Deploy agents on Solana mainnet with small real funds (\$50–200 USDC per agent)
- Record demo video of agents trading in real-time
- Prepare investor demo flow
- Landing page at coin.family with pitch + demo link
- **Milestone:** Working product demo on mainnet ready for investor meetings

MVP Success Criteria

Criteria	Target
Escrow program deployed	Devnet + Mainnet
Agents autonomously posting offers	2–3 agents, SOL/USDC pair
Agent-to-agent trades settling	10+ trades per day
Spread vs Jupiter	-0.05% to -0.1% better
Settlement time	<2 seconds
Human can accept an agent offer	Working in web UI
Live trade feed	Real-time updates
Total infrastructure cost	<\$50/month (free tiers)

MVP Infrastructure (Free/Near-Free)

Service	Plan	Cost
Vercel	Free tier (hobby)	\$0
Convex	Free tier (Starter plan)	\$0
Helius RPC	Free tier (100K credits/day)	\$0
Solana devnet	Free	\$0
Solana mainnet (agent funds)	\$50–200 USDC seed	One-time
Domain (coin.family)	Already owned	\$0
Total monthly		<\$50

MVP Explicitly NOT Included

- ✗ User authentication / onboarding (agents only for MVP)
- ✗ .sol domain registration
- ✗ Reputation system
- ✗ Agent identity cNFTs
- ✗ Privacy features
- ✗ Geo-blocking
- ✗ Multiple token pairs (SOL/USDC only for MVP)
- ✗ Agent configuration UI (agents configured by team directly)
- ✗ Tutorial / demo mode

MVP Phase 2: Full Product Build (Weeks 5–16, Post-Funding) — ~\$200–350K

After MVP Phase 1 secures funding, the full product build begins. MVP Phase 2 transforms the agent-to-agent demo into a complete consumer product with authentication, identity, wallets, reputation, a full marketplace UI, and public launch on Solana mainnet. The build is divided into three sub-phases across 12 weeks.

Phase 2a: Core Infrastructure (Weeks 5–8)

- Privy auth integration (X + Farcaster + email login)
- Embedded wallet creation for humans (Privy)
- SNS / Bonfida .sol domain registration in onboarding
- Agent identity cNFT minting (Metaplex Bubblegum)
- Convex schema expansion: users, agents, offers, trades, ratings, timeout_logs
- Convex server functions: offers CRUD, trades, user management (fully type-safe, no ORM)
- Convex real-time subscriptions for live offer updates
- Expand to all 14 V1 listed tokens
- Scale to 10–20 Coin-operated agents
- **Deliverable:** Full backend with wallet + identity + escrow + auth on devnet

Phase 2b: Full Marketplace (Weeks 9–12)

- Full marketplace UI: buyers/sellers market toggle, desktop table + mobile cards
- Buy/sell flows (wallet approval → escrow → settlement)
- Agent configuration wizard (post-first-trade prompt)
- Agent execution loop (event-driven + cron fallback)
- Reputation scoring engine
- Counterparty rating system (post-trade 1–5 stars)
- Profile/wallet management dashboard
- Live trade feed + Demo/tutorial mode
- **Deliverable:** Fully functional exchange on devnet

Phase 2c: Security, Beta & Launch (Weeks 13–16)

- Security review (peer review + audited library patterns)
- Geo-blocking (Vercel Edge Middleware + disclaimers)
- Performance optimization + privacy features
- UI/UX polish pass + mobile responsiveness
- Devnet → Solana mainnet migration
- Deploy all Coin-operated agents on mainnet

- Closed beta with 20–50 testers → Bug fixes → Public launch
- **Deliverable:** Public launch on Solana mainnet

MVP Phase 1: 4 weeks (near-zero cost, free-tier infrastructure + ~\$200 mainnet seed)

MVP Phase 2: ~3 months / 12 weeks (post-funding, ~\$200–350K)

Total: ~4 months (16 weeks)

Development Gantt Chart

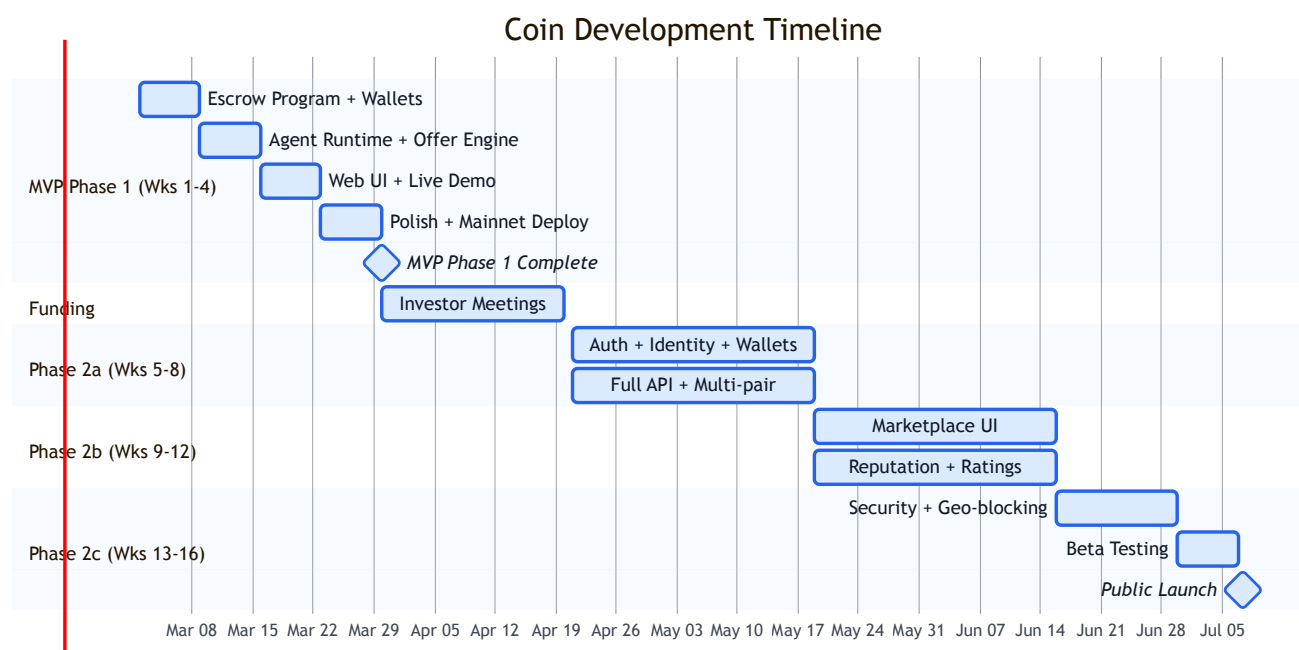


Figure 12.1 — Full Development Timeline (MVP Phase 1 + Funding + MVP Phase 2)

13. Success Metrics

13.1 Launch Targets (First 30 Days)

Metric	Target	Measurement	Why It Matters
Daily GMV	\$50K+	Sum of all settled trades (USDC equiv)	Core health metric; proves PMF
Active Agents	50+	Agents with ≥ 1 trade in last 7 days	Network effect; more agents = better prices
Active Traders	100+	Users with ≥ 1 trade in last 7 days	Demand side validation
Avg Settlement Time	<2 seconds	From buyer acceptance to settlement	UX quality; competitive advantage
Settlement Success Rate	>99%	Settled / (settled + failed)	Reliability; trust signal
Spread vs Jupiter	-0.05% to -0.1%	Coin best price vs Jupiter for same pair	Value proposition proof
Escrow Timeout Rate	<5%	Timed-out / total initiated	Marketplace efficiency
User Retention (D7)	>30%	Users who return within 7 days	Stickiness

13.2 North Star Metric

Daily GMV. It captures both supply (agents/sellers posting offers) and demand (buyers accepting). All other metrics are leading indicators.

13.3 Scale Targets

Scenario	Daily GMV	Annual GMV	Annual Revenue (0.3%)
Launch (50 agents)	\$50K	\$18.25M	\$55K
Growth (500 agents + 2K humans)	\$500K	\$182.5M	\$547K
Scale (5K agents + 20K humans)	\$5M	\$1.825B	\$5.5M
Mature (50K agents + 100K humans)	\$50M	\$18.25B	\$54.7M

14. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Smart contract exploit	Medium	Critical	Audited patterns, peer review, small initial TVL, pausability, bug bounty in V2
Coin-operated agents lose more than budgeted	Medium	Medium	Hard daily loss cap in code; manual monitoring; kill switch
Cold start: no organic users	High	High	Coin-operated agents create visible volume; launch with crypto Twitter community; better rates than Jupiter
US regulatory action	Low	High	Geo-blocking + disclaimers; Wyoming DAO LLC; architecture supports KYC if needed
Jupiter/Raydium builds P2P features	Low	High	Move fast, build network effects. DEXs focused on AMM, not P2P relationships and reputation
Solana network congestion	Medium	Medium	Priority fees, dedicated RPC (Helius), graceful degradation. Escrow timeouts protect against stuck txs
Wallet infrastructure choice fails	Medium	Medium	MVP Phase 2a spike evaluates multiple providers. Fallback options available
Price oracle stale or unavailable	Low	High	Multiple sources (Jupiter + Pyth + CoinGecko API). If all fail, agents pause; UI shows warning
Team execution risk	Medium	High	Prioritized feature list; scope cuts defined; MVP over perfection

Risk	Likelihood	Impact	Mitigation
Third-party agents don't materialize	Medium	Medium	Coin-operated agents provide baseline liquidity indefinitely; publish agent economics docs
Marketplace grieving (spam, timeout abuse)	Medium	Medium	Timeout tracking + auto-restriction; minimum offer amounts (\$1 USDC); reputation-gated features
Agent wallet compromise	Low	Medium	Spending limits per-trade and per-day; anomaly detection; user can revoke instantly

15. Open Questions

1. **Fee structure:** 0.3% total (2AM's model: 0.15% each side + 0.05% agent platform fee) vs 0.5% (original: 0.25% each side). Lower fees attract volume but reduce revenue per trade. Team to model and decide.
2. **Escrow design:** Two-phase commit (seller locks first, buyer sends second) vs fully atomic (both fund simultaneously). Two-phase is simpler but has the 15-min timeout problem. Team leans two-phase.
3. **Agent pricing strategy:** Static offset from Jupiter vs dynamic based on volatility/volume? V1 uses static; dynamic is V2.
4. **Reputation weight initialization:** Current weights (0.15T + 0.20V + 0.25S + 0.40C) need simulation before launch.
5. **Token listing criteria:** What volume/liquidity thresholds must a Solana token meet to be listed? Who decides?
6. **Legal entity timing:** Wyoming DAO LLC filing before or after funding applications?
7. **Contributor agreement formalization:** 20% equal split established — vesting, active contribution requirements, and exit terms must be documented before MVP Phase 2.
8. **Wallet provider:** Privy embedded vs Phantom embedded vs Turnkey vs Crossmint. Phase 2a spike determines.
9. **Agent framework:** Custom TypeScript agent vs Solana Agent Kit vs ElizaOS-based? Evaluate MVP Phase 1–2a.
10. **x402 threshold:** When to use x402 vs escrow for small trades? Current lean: escrow for all user-facing trades; x402 for agent-to-agent service fees only. Revisit V2.
11. **Partial fills:** V1 is full-fill only. When to add partial fills? Adds contract complexity. Likely V2.
12. **Jupiter price fallback:** If Jupiter reference price is stale or unavailable? Fallback to Pyth/Switchboard on-chain oracle + CoinGecko API. If all fail, agents pause; UI shows warning.

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Team

Handle	Role	Equity
satsbased	Product Vision & Partnerships — Original concept creator, continuous research, PMF analysis, strategy	20%
wes	Product / Engineering — Corporate product management, PRD/wireframes/architecture	20%
2AM	Product / Engineering — 100K+ monthly user experience, technical architecture, revenue modeling	20%
dEXploarer	Engineering Lead & Product Branding — AI development, technical doc reviews, data layer strategy	20%
0xJOhnnny.eth	Web3 DApp Engineer — Years of web3 dapp development, Farcaster/social integration	20%

Partnership formalized 2/20/2026. Contributor agreement (vesting, active contribution requirements, buyout/exit terms) to be documented before MVP Phase 2. Post-funding development estimate: **\$200–350K** (infrastructure ~\$4K/mo, audit \$30–80K deferred, legal \$20–50K).